



JKBOSE 2026

CLASS 11TH

CONIC SECTIONS

MATHS

LECTURE-4

BY-MEHRAJ SIR



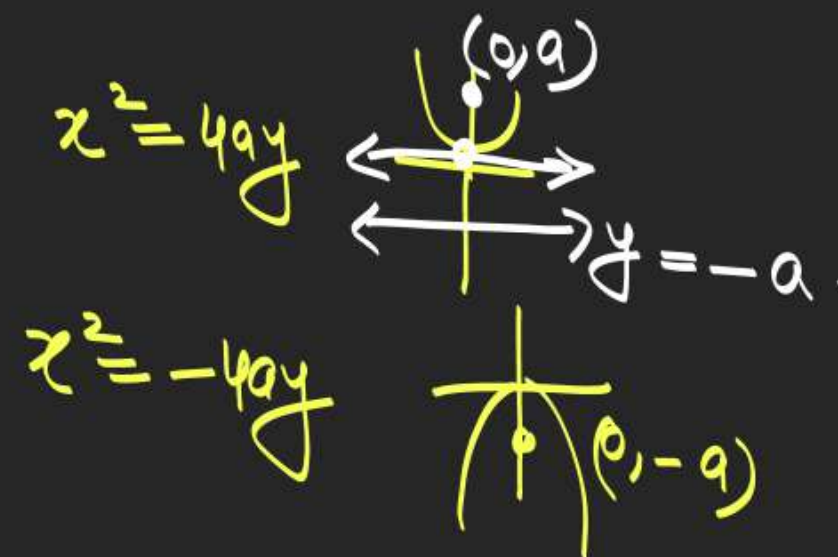
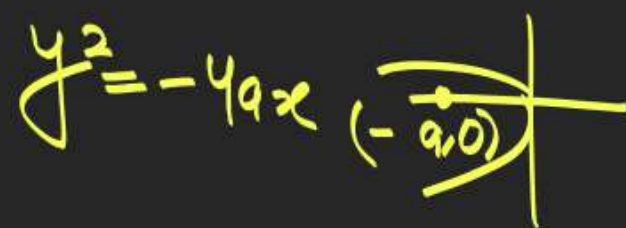
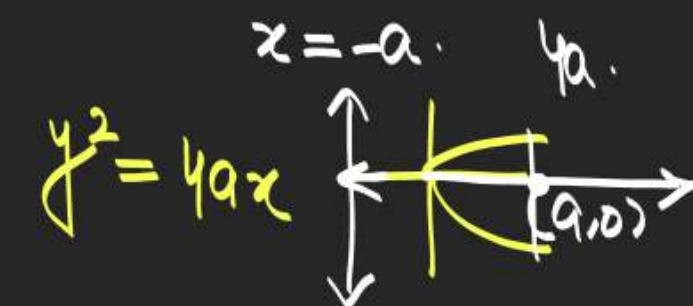
TOPICS TO BE COVERED

1

Exercise Discussion 10.2

2

Related Problems





— WELCOME TO —
JKBOSE

MISSION RANKERS



TODAY IS THE

1ST CLASS

OF JKBOSE MISSION RANKERS



CLASS TIME:

8:00 PM



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JKBOSE WALLAH



JKBOSE WALLAH 9TH & 10TH

EXERCISE – 10.2

QUESTION

In each of the following Exercises 1 to 6, find the coordinates of the focus, axis of the parabola, the equation of the directrix and the length of the latus rectum.

$$y^2 = 12x$$

(Sol) Given Parabola is:

$y^2 = 12x = 4(3)x$, which is clearly a Right handed parabola. So,
Focus = $(3, 0)$

Axis of parabola — x -axis.

Directrix : $x = -3$ or $x + 3 = 0$.

$$\begin{aligned} \text{Length of latus Rectum} &= 4a \\ &= 4 \times 3 \\ &= 12. \end{aligned}$$

Hence the result.

QUESTION

In each of the following Exercises 1 to 6, find the coordinates of the focus, axis of the parabola, the equation of the directrix and the length of the latus rectum.

$$x^2 = 6y$$

(Sol) Given Parabola is:

$$x^2 = 6y = 4\left(\frac{3}{2}\right)y$$

$\Rightarrow x^2 = 4\left(\frac{3}{2}\right)y$, which is clearly an upward parabola.

So, focus = $\left(0, \frac{3}{2}\right)$

axis of Parabola — y-axis.

Directrix: $y = -\frac{3}{2}$ or $2y + 3 = 0$.

Length of latus Rectum

$$= 4 \times \frac{3}{2}$$

$$= 6$$

Hence the result.

QUESTION

In each of the following Exercises 1 to 6, find the coordinates of the focus, axis of the parabola, the equation of the directrix and the length of the latus rectum.

$$y^2 = -8x \quad (-a, 0)$$

HP/W

QUESTION

In each of the following Exercises 1 to 6, find the coordinates of the focus, axis of the parabola, the equation of the directrix and the length of the latus rectum.

$$x^2 = -16y$$

H/W

QUESTION

In each of the following Exercises 1 to 6, find the coordinates of the focus, axis of the parabola, the equation of the directrix and the length of the latus rectum.

$$y^2 = 10x \quad \text{H/W.}$$

QUESTION

In each of the following Exercises 1 to 6, find the coordinates of the focus, axis of the parabola, the equation of the directrix and the length of the latus rectum.

$$x^2 = -9y$$

H/W

QUESTION

Focus $(6, 0)$; directrix $x = -6$

(Sol) Here Focus is $(6, 0)$. So $a = 6$.

$$\Rightarrow y^2 = 4 \times 6x$$

$\Rightarrow y^2 = 24x$ is the required parabola.

QUESTION

Focus $(0, -3)$; directrix $y = 3$

(Sol) Here focus $(0, -3)$ & directrix is $y = 3$

$$\Rightarrow x^2 = -4 \times 3y$$

$\Rightarrow x^2 = -12y$ is the required parabola.

QUESTION



Vertex $(0, 0)$; focus $(3, 0)$

$$y^2 = 4ax$$

$$\Rightarrow y^2 = 12x$$

QUESTION

Vertex $(0, 0)$; focus $(-2, 0)$

- (a) Right handed Parabola. ✗
- (b) left handed parabola. ✓
- (c) Upward Parabola. ✗
- (d) Downward Parabola. ✗

$$y^2 = -4 \times 2x$$
$$\Rightarrow y^2 = -8x.$$

QUESTION

Vertex $(0, 0)$ passing through $(2, 3)$ and axis is along x -axis.

(Sol) Suppose that $y^2 = 4ax$ — (1) be the given parabola

Since $(2, 3)$ lies on (1), then

$$3^2 = 4a \times 2$$

$$\Rightarrow a = \frac{9}{8} \text{ — (2)}$$

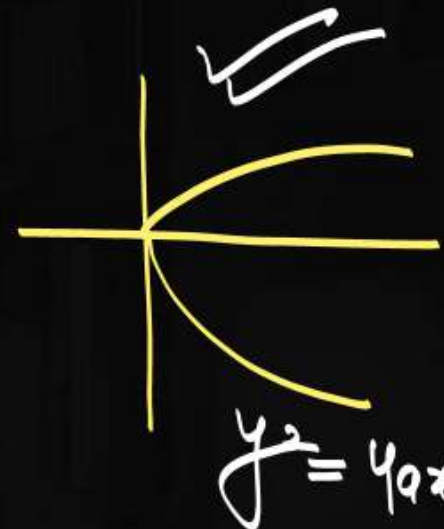
Using (2) in (1), we get;

$$y^2 = 4\left(\frac{9}{8}\right)x$$

$$\Rightarrow 2y^2 = 9x$$

$\Rightarrow 2y^2 - 9x = 0$ is the required parabola.

Hence the result.



QUESTION

Vertex (0, 0), passing through (5, 2) and symmetric with respect to y-axis.

(Sol) Suppose that $x^2 = 4ay$ — (1) be the given parabola.

Since (1) passes through (5, 2). Then $\Rightarrow 2x^2 - 25y = 0$ which

$$5^2 = 4a \times 2.$$

$$\Rightarrow a = \frac{25}{8} \text{ — (2)}$$

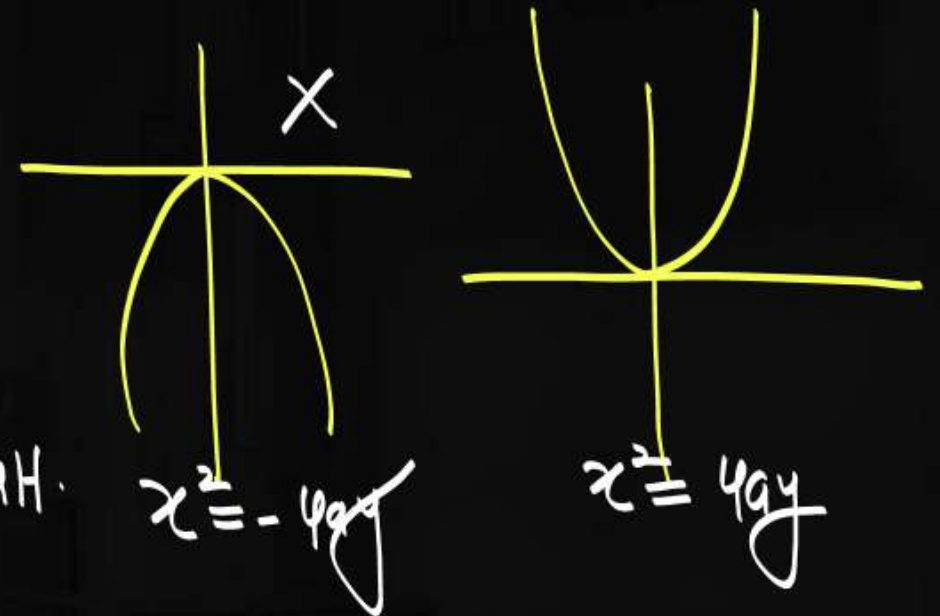
Using (2) in (1), we get;

$$x^2 = 4 \left(\frac{25}{8} \right) y$$

$$\Rightarrow 2x^2 = 25y$$

is the required parabola.

Hence the result.



QUESTION (NCERT EXAMPLER):

The centre of the circle passing through the point $(0, 1)$ and touching the parabola $y = x^2$ at the point $(2, 4)$ is: (Slope of Parabola at $(2, 4)$ is 4)

1 $\left(\frac{3}{10}, \frac{16}{5}\right)$

(Sol) clearly $OP = OQ$ & $OP \perp l$
 $\Rightarrow \sqrt{(h-2)^2 + (k-4)^2} = \sqrt{h^2 + (k-1)^2}$
 $\Rightarrow h^2 + 4 - 4h + k^2 + 16 - 8k = h^2 + k^2 - 2k + 1$

2 $\left(-\frac{16}{5}, \frac{53}{10}\right)$

$\Rightarrow -4h - 8k + 20 = -2k + 1$
 $\Rightarrow -4h - 6k = -19$

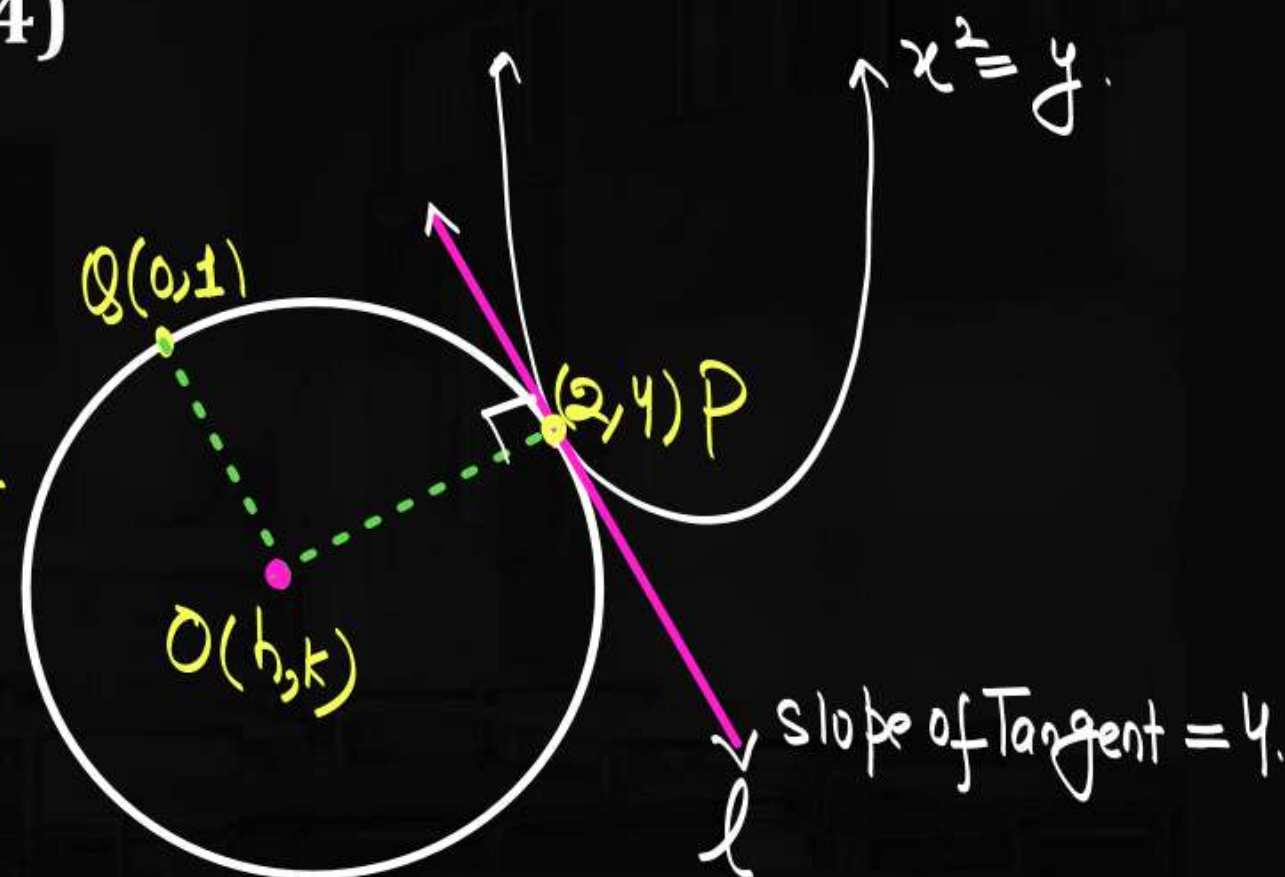
3 $\left(\frac{6}{5}, \frac{53}{10}\right)$

$\Rightarrow 4h + 6k = 19$ — (1)

4 $\left(-\frac{53}{10}, \frac{16}{5}\right)$

Also $OP \perp l$

$\Rightarrow \frac{k-4}{h-2} \times 4 = -1$



$$\Rightarrow \frac{k-4}{h-2} = -\frac{1}{4}$$

$$\Rightarrow k-4 = -\frac{1}{4}(h-2)$$

$$\Rightarrow k-4 = -\frac{h}{4} + \frac{2}{4}$$

$$\Rightarrow k-4 = \frac{1}{2} - \frac{h}{4} = \frac{2-h}{4}$$

$$\Rightarrow k = \frac{2-h}{4} + 4$$

$$\Rightarrow \boxed{k = \frac{18-h}{4}} \text{--- (2)}$$

Using (2) in (1), we get;

$$4h + 3\left(\frac{18-h}{4}\right) = 19$$

$$\Rightarrow 4h + \frac{3(18-h)}{4} = 19$$

$$\Rightarrow 8h + 54 - 3h = 38$$

$$\Rightarrow 5h = -16$$

$$\Rightarrow \boxed{h = -\frac{16}{5}}$$

$$\text{So, } k = \frac{18 + \frac{16}{5}}{4} = \frac{\frac{106}{5}}{4}$$

$$\Rightarrow \boxed{k = \frac{53}{10}}$$

So, the required Centre

$$= (h, k)$$

$$= \left(-\frac{16}{5}, \frac{53}{10}\right)$$

QUESTION (JEE-MAINS 2020)

If $y = mx + 4$ is a tangent to the parabola, $y^2 = 4x$ and $x^2 = 4y$, then b is equal to :

1 -64

2 -32

3 -128

4 128

★ Condition for tangency:

Suppose that $y^2 = 4ax$ — (1) be the given parabola

& let $y = mx + c$ — (2) be the Tangent to (1).

Using (2) in (1), we get

$$(mx + c)^2 = 4ax$$

$$\Rightarrow m^2x^2 + c^2 + 2mcx = 4ax$$

$$\Rightarrow m^2x^2 + (2mc - 4a)x + c^2 = 0 \text{ — (3)}$$

Since (2) is a Tangent to (1), so the roots of the equation (3) are real & equal.

$$\Rightarrow (2mc - 4a)^2 - 4m^2c^2 = 0$$

$$\Rightarrow \cancel{4m^2c^2} + 16a^2 - 16mca - \cancel{4m^2c^2} = 0$$

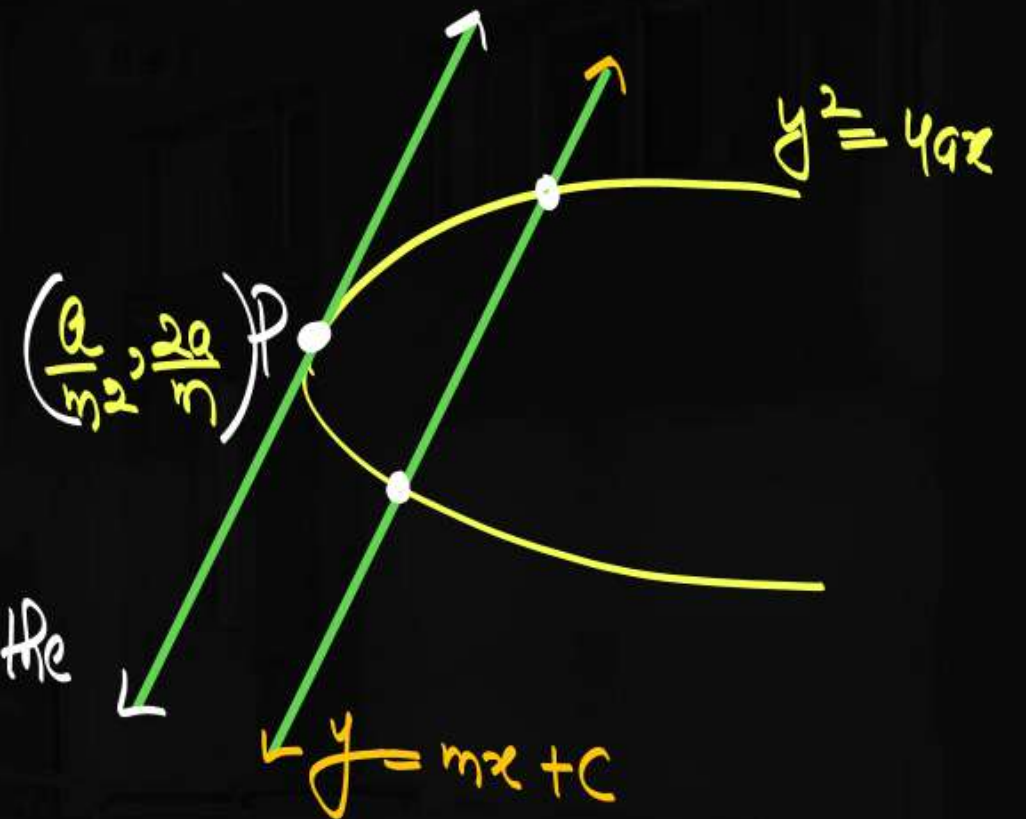
$$\Rightarrow \cancel{16a^2} = \cancel{16mca}$$

$$\Rightarrow mc = a$$

$$\Rightarrow \boxed{c = \frac{a}{m}} \text{ which is the required Condition.}$$

Also the equation of Tangent to (1) is;

$$\boxed{y = mx + \frac{a}{m}} \text{ — (4)}$$



Point of Contact:

Using ④ in ①, we get;

$$\left(mx + \frac{a}{m}\right)^2 = 4ax$$

$$\Rightarrow m^2x^2 + \frac{a^2}{m^2} + 2\cancel{mx} \times \cancel{\frac{a}{m}} = 4ax$$

$$\Rightarrow m^2x^2 + \frac{a^2}{m^2} + 2ax - 4ax = 0$$

$$\Rightarrow m^2x^2 + \frac{a^2}{m^2} - 2ax = 0$$

$$\Rightarrow \left(mx - \frac{a}{m}\right)^2 = 0$$

$$\Rightarrow mx - \frac{a}{m} = 0$$

$$\Rightarrow \boxed{x = \frac{a}{m^2}}$$

$$\text{So, } y = \cancel{\left(\frac{a}{m^2}\right)} + \frac{a}{m}$$

$$\Rightarrow y = \frac{a}{m} + \frac{a}{m} = \frac{2a}{m}$$

$$\text{So, Point of Contact} = \left(\frac{a}{m^2}, \frac{2a}{m}\right)$$

QUESTION (JEE-MAINS 2020)

If the common tangent to the Parabolas $y^2 = 4x$ and $x^2 = 4y$ also touches the circle, $x^2 + y^2 = c^2$, then c is equal to :

~~1~~ $\frac{1}{2}$

~~2~~ $\frac{1}{4}$

~~3~~ $\frac{1}{2\sqrt{2}}$

~~4~~ $\frac{1}{\sqrt{2}}$

(Sol) Given Parabolas are:

$$y^2 = 4x \text{ --- (1)}$$

$$x^2 = 4y \text{ --- (2)}$$

Suppose that $y = mx + \frac{1}{m}$ --- (3) be the

Tangent (1) & (2)

Using (3) in (2), we get:

$$x^2 = 4\left(mx + \frac{1}{m}\right) = 4mx + \frac{4}{m}$$

$$\Rightarrow mx^2 - 4m^2x - 4 = 0 \text{ --- (4)}$$

Since (3) is a Tangent to (1), so the roots of (4) are real & equal.

$$\Rightarrow (-4m^2)^2 - 4m(-4) = 0$$

$$\Rightarrow 16m^4 + 16m = 0$$

$$\Rightarrow m^3 + 1 = 0$$

$$\Rightarrow m = -1$$

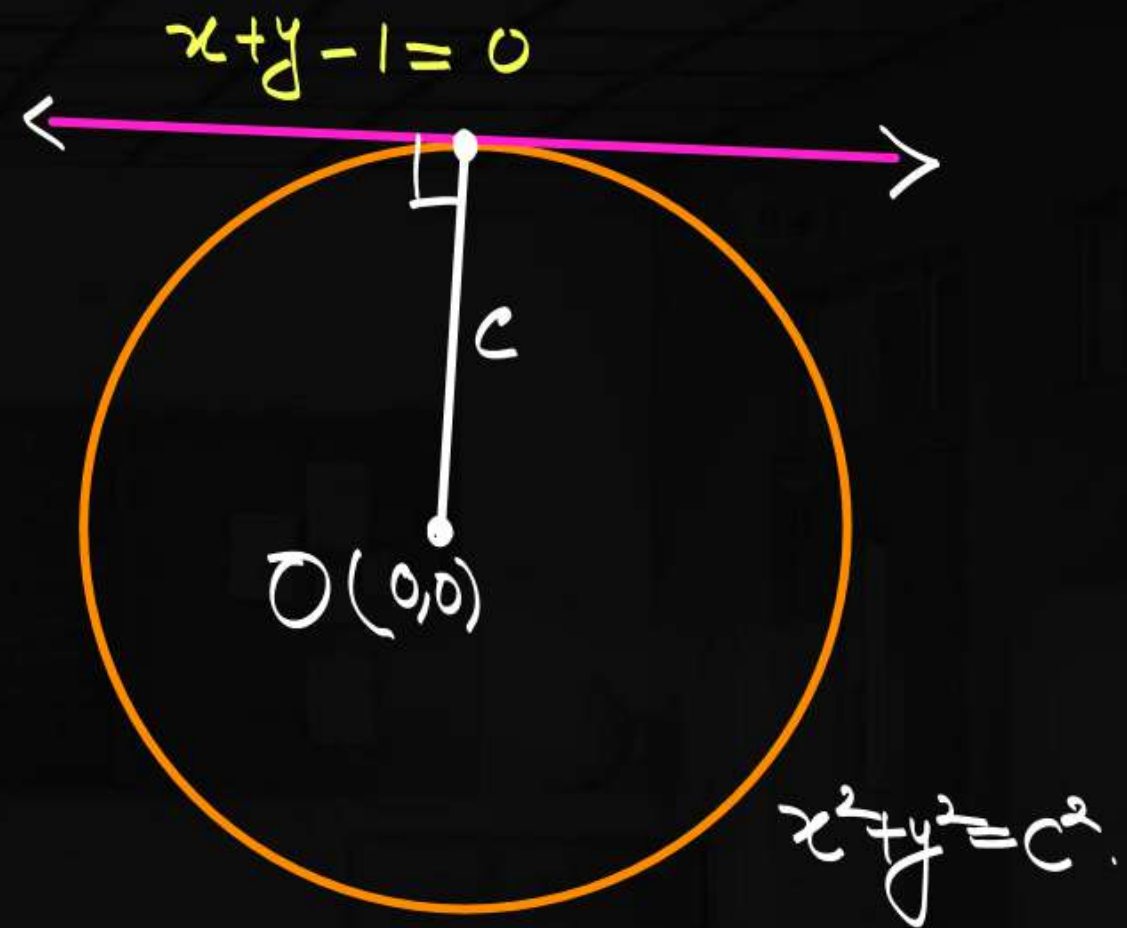
So, the required Common Tangent of (1) & (2) is:

$$y = -x + 1$$

$$\Rightarrow x + y - 1 = 0$$

Here $c = \frac{|0+0-1|}{\sqrt{1^2+1^2}}$

$\Rightarrow c = \frac{1}{\sqrt{2}}$



QUESTION (JEE-MAINS 2020)

The area(in sq.units) of an equilateral triangle inscribed in the parabola $y^2 = 8x$, with one of its vertices on the vertex of this Parabola, is :

(Homework Problem)

1 $64\sqrt{3}$

2 $256\sqrt{3}$

3 $128\sqrt{3}$

4 $192\sqrt{3}$

Tawkeea Amin safi 

Rasib Ayoub Rather 

Firdousa Majeed 

Thank
You